HAB PAYLOAD STABILIZATION SYSTEM

Inverted Inertia Platform — As-Built State & Upgrade Roadmap

K6ATV | *June 2026* | *Revision 1.1*

Design Summary: A HAB payload pointing system using a gimbal BLDC motor and four 500 mm carbon-fibre arms to orient a passive 800 g camera payload. No slip rings, no swashplate, no saturation condition, no active electronics on the payload. Payload heading is derived from an IMU on the arm plus a magnetic encoder reading a passive magnet — nothing electrical crosses the rotating interface.

Status — June 2026 launch & encoder upgrade in progress

This revision documents the system **as actually built and flown**. The AS5048A magnetic encoder is presently read in **PWM single-wire mode** (the cable that shipped pre-installed with the motor). The encoder hardware also supports SPI, and the **next planned upgrade is to move the encoder to SPI** to make rotation smoother — see “Known Issue / Next Upgrade” in Section 7.

1. How It Works

1.1 Inverted Architecture

Rather than mounting a reaction wheel on the payload, this design inverts the concept. The arm assembly is the active control platform — it drives the payload to the correct heading. The GBM2804H gimbal outrunner motor connects the two assemblies: stator fixed to the arm, rotor bell fixed to the payload top plate. A 35 mm bearing carries all structural loads independently of the motor.

The GBM2804H is a **gimbal-wound BLDC**: a low-Kv (154 Kv), high pole-count machine (12N14P, 7 pole pairs) with relatively high phase resistance (~10 Ω). Unlike a high-Kv drone motor built to spin fast, a gimbal motor is designed for smooth, high-torque positioning at very low RPM — which is exactly the regime this platform operates in, holding and nudging payload heading rather than rotating continuously. That low-RPM positioning duty is also what makes precise encoder feedback critical (see Section 7).

1.2 Heading Sensing — Zero Payload Electronics

Payload absolute heading is computed entirely from sensors on the arm. The BNO085 provides the arm absolute world-frame yaw. The AS5048A (integrated into the motor) reads a magnet glued to the payload top plate. Summing them gives payload absolute heading across unlimited rotations — firmware tracks full turns to prevent wrap-around. The payload contributes only a passive N52 magnet (~1 g).

1.3 No Saturation — Positioning Actuator

A conventional reaction wheel saturates by accumulating angular momentum until it hits maximum RPM. This design has no saturation: the motor is a positioning actuator. When the payload drifts, the motor drives it back. Nothing accumulates. Unlimited corrections over the full flight at no cost to authority.

1.4 The r² Inertia Advantage

Placing electronics at 300–500 mm radius instead of a compact central flywheel gives a large inertia advantage. At 800 g payload the inertia ratio is about 25× — the arm resists corrections ~25 times more than the payload rotates, so motor corrections barely disturb the arm in normal operation.

Configuration	Mass	Radius	Inertia	Ratio
Compact 100 mm flywheel	120 g	50 mm	1.5e-4 kg·m²	1×
Arm electronics (avg 400 mm)	300 g	~400 mm	0.048 kg·m²	320×
Full arm assembly	323 g	various	~0.040 kg·m²	~267×

2. Firmware — Control Loop

Heading is computed at 100 Hz on the Feather STM32F405. payload_hdg = arm_yaw (BNO085) + motor_angle (AS5048A). A PID loop (Kp=0.50 Ki=0.02 Kd=0.08) drives the shortest-path angular error to zero, with a 1.0° dead-band to avoid correcting trivial errors. The SimpleFOC inner current loop runs at 1000 Hz. Above 15 000 m the IMU switches to gyro-dominant heading (magnetometer unreliable at altitude).

3. Component Specification (Arm Assembly)

Component	Part / Specification	Mass	Notes
Gimbal BLDC Motor + Encoder	iPower GBM2804H-100T (gimbal-wound) w/ integrated AS5048A; 12N14P, 10 Ω, 154 Kv	51 g	Low-RPM positioning; encoder in PWM mode (see §7)
Motor Driver	SimpleFOC Mini v1.1; DRV8313, 3-PWM, 5A max	8 g	Onboard 3.3V regulator
Microcontroller	Feather STM32F405 Express; 168 MHz M4F	5 g	Logic from Mini 3.3V; never 9V to BAT
IMU	Adafruit BNO085 #4754; onboard fusion	3 g	I2C addr 0x4A; gyro-only >15 km
Bearing	35 mm ID deep groove, stainless	20 g	Carries all structural loads
Arms ×4	6 mm OD 1 mm wall CF tube, 500 mm, 90°	128 g	32 g each; cross-pattern
Battery	6× Energizer L91 AA, series, 9V, 3000 mAh	87 g	Rated -40C; ~4.3 h cold-derated
Frame / Wiring	PCB mounts, connectors, harness	30 g	Estimate

Arm assembly total: ~332 g.

4. Electrical — Feather STM32F405 Pin Mapping

Pin assignments below match **src/config.h** as-built. The three motor IN pins are all on hardware timer channels; EN is any digital pin. The AS5048A is currently a single PWM line (not SPI — see Section 7).

Signal	Feather Pin	Bus / Mode	Notes
SimpleFOC IN1	D9 (PB8, TIM4_CH3)	PWM	Motor phase A
SimpleFOC IN2	D10 (PB9, TIM4_CH4)	PWM	Motor phase B
SimpleFOC IN3	D5 (PC7, TIM3_CH2)	PWM	Motor phase C
SimpleFOC EN	D6 (PC6)	Digital	Driver enable

AS5048A angle	D11 (PC3) — PWM single-wire	PWM in	MagneticSensorPWM; SPI is the planned upgrade
BNO085 SDA/SCL	PB7 / PB6 (STEMMA QT)	I2C1	addr 0x4A
Battery sense	A0 (PA4)	ADC	10k/3.3k divider, ratio 4.03
Motor + logic supply	Mini VIN ← 9V; Feather 3V ← Mini 3.3V	Power	Never connect 9V to Feather BAT/USB

Telemetry: USB CDC serial, 115200 baud, every 30 s. Battery warn 7.8 V, cutoff 7.2 V; motor voltage limit 7.0 V from a 9.0 V nominal supply.

5. Performance Summary

Parameter	Value	Notes
Designed payload mass	800 g	Camera + enclosure + tracker
Maximum payload mass	1.2 kg	10× inertia margin maintained
Inertia ratio at 800 g	25×	Arm 25× harder to spin than payload
Encoder resolution	0.022°	AS5048A 14-bit (sensor limit)
Motor max torque	~56 mN·m	GBM2804H at 9V (peak)
Saturation risk	None	Positioning actuator
Flight duration covered	2.8 h to 30 km	Battery gives 4.3 h cold-derated
Battery chemistry	Li-FeS2 (L91)	Rated -40C, functional to ~-55C
Rotating electrical conn.	Zero	No slip rings anywhere
Payload active electronics	Zero	Passive magnet only

6. Key Design Decisions

Finless design

At 800 g payload the arm inertia ratio is 25× without fins. Ascending airflow creates velocity-dependent drag on bare CF arms opposing rotation. Fins can be retrofitted if testing shows excessive arm spin.

Motor with integrated AS5048A encoder

The GBM2804H-100T with integrated AS5048A eliminates manual encoder alignment — the factory centres the magnet on the shaft axis. 14-bit resolution improves FOC smoothness at the near-zero RPM this system runs at.

6× AA Energizer L91 in series (9V)

Rated -40C, functional to ~-55C float altitude. 3000 mAh gives ~4.3 h cold-derated runtime vs ~2.8 h flight. LiPo cells degrade severely at float altitude; L91 cost ~$10 per flight, 87 g total.

Bearing separate from motor load path

The 35 mm bearing carries all payload weight and pendulum swing forces. The motor handles only rotational torque, keeping it in its designed torque-only regime.

7. Known Issue / Next Upgrade — Encoder PWM → SPI

Lesson learned — June 2026 launch

The AS5048A shipped with a pre-installed **PWM output cable**, so the encoder was wired and flown in PWM mode to save bring-up time. Bench testing of the PWM readings looked accurate enough at a glance, so it appeared safe to fly that way.

In flight / testing this proved insufficient. A gimbal motor run as a fine-positioning actuator at near-zero RPM needs very exact, low-latency angle data to commutate smoothly. The PWM single-wire decode adds enough timing jitter and latency that the control loop exhibited twitching and related instability — static readings can look fine while the loop still lacks the precision it needs.

Next upgrade: move the AS5048A to its native **SPI** interface (MISO/SCK/MOSI/CSN on SPI2), replacing MagneticSensorPWM with MagneticSensorSPI in firmware. SPI gives the deterministic, high-resolution angle reads the FOC loop needs and is expected to substantially smooth rotation. This is the primary planned change before the next flight.

8. Remaining Work Before Next Flight

Area	Task	Priority
Firmware	Convert encoder PWM → SPI (MagneticSensorSPI, SPI2 + CSN)	High
Firmware	Re-verify encoder sign convention on bench after SPI swap	High
Firmware	Re-tune PID with SPI encoder (start Kp=0.5 Ki=0.02 Kd=0.08)	High
Mechanical	Arm-to-rotor-bell interface plate + AS5048A magnet mount	High
Mechanical	Verify arm mass balance (asymmetry causes pendulum oscillation)	High
Firmware	Add UART altitude input from tracker for mag cutoff	Medium
Thermal	Freezer test BNO085 and AS5048A at -20C; characterise drift	High
Testing	Bench: hold payload heading while manually rotating arm rig	High

As-built + roadmap — K6ATV — June 2026. github.com/radiohound — Specifications subject to revision following the encoder SPI conversion and prototype testing.